DS4021 - Final Project Report
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FINAL PROJECT REPORT: Predicting NYC Restaurant Health Inspections

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Introduction

Research Question: Can we classify New York City restaurants into health risk categories (Grades A, B, or C) based on features e.g. location, type of cuisine, borough, etc, to proactively identify potential health code violations?

Foodborne illness is a significant public health challenge which affects millions of individuals every year. To mitigate such risks, New York City employs a letter-grade safety system which is typically displayed in restaurant windows to inform potential customers of that restaurant's hygiene standards. However, with thousands of diverse establishments across the five boroughs, inspection resources are limited. Identifying which factors and features are the strongest predictors of health code violations is critical. Understanding these relationships may enable health departments to utilize predictive models to help allocate limited inspection resources more efficiently by identifying higher-risk establishments before they pose a significant public health threat. The primary goal of this project is to develop and evaluate a machine learning model to predict the likelihood of a New York City restaurant receiving a critical violation grade.

Methods

The dataset chosen is the "NYC Restaurant Inspection Results" dataset from NYC Open Data, consistenting of about 200,000 inspection records. Key features selected for analysis included borough, cuisine type, zipcode, and latitude/longitude. The main challenge was that the dataset included two metrics which went hand in hand: grade and score. In the NYC system, the 'Grade' column is mathematically derived from the numeric 'Score' column (ex. 0-13 points =A). To prevent target leakage, which would be the model effectively memorizing the translation from 'Score' to 'Grade', we removed the 'Score' column from our feature set. Another factor to consider was that there was a large skew in the data with the large majority of grades, roughly 90%, being As. We chose to evaluate all of the models' Macro F1 score within the cross validation loops, as a way to evaluate the multi-class performance given this large imbalance of grade distribution.

We trained and optimized four classifier models: a penalized linear model, a support vector machine (LinearSVC), ensemble model (Random Forest), and a neural network. To ensure proper data integrity, we used pipelines imported from 'sklearn.pipeline.Pipeline' to ensure scaling numerical data and one-hot encoding within the cross-validation loop. To gauge accuracy for models, we used Macro F1 score for assessing the regression models, while using accuracy and confusion matrices for evaluating classification models.

Results

Our evaluation of the four models revealed distinct differences among their prediction accuracy/ capabilities. The penalized linear model and neural network failed to capture any significant underlying patterns, resulting in a F1 score value near zero 0.132. While the Linear Support Vector Classifier (LinearSVC) appeared stable with a cross-validation accuracy of about 54%, it demonstrated weakness in detecting minority classes such as Grades B and C. The Linear SVC heavily favored the majority class ('A'), failing to detect high-risk restaurants (Grade 'B' Recall: 0.05, Macro F1: 0.23). In contrast, the Ensemble (Random Forest) model performed better with a cross-validation macro F1 score of 0.550. On the withheld test set, the Random Forest achieved a macro F1 score of 0.562, a bit higher than the same model on the training set.

Model	**Task Type**	**Primary Metric**	**CV Score**	**Notes**
Ensemble (Random Forest)	Classification	Macro F1	0.562	Best Performance: 2x better at detecting risk than SVM
Linear SVC	Classification	Macro F1	0.230	Stable accuracy of about 55%, but failed to recall minority classes (Grades 'B' and 'C')
Penalized Linear Model	Classification	Macro F1	0.136	Linear assumption unfit for complex non-linear data
Neural Network	Classification	Macro F1	0.129	Failed to converge or capture patterns

Discussion

Our Random Forest model achieved a test accuracy of about 80.2% with a macro F1 score of about 65%. Although the random forest model displayed some signs of overfitting with a training accuracy of 87% against 56% cv score, it was the only model capable of distinguishing between passing and failing grades. As a result, we selected the Random Forest model to be the best performing model. One of the main struggles seemed to be that many of the models were not able to distinguish 'B' and 'C' grades from 'A' since 'A' had a large distribution within the original dataset. Even after implementing the class_weight to be balanced within the LinearSVC, it still had a low Macro F1 score of 0.23. It seems that Random Forest was able to handle these distinctions the best, leading to a stronger performance. A major limitation of this project is the strong class imbalance in the dataset, with nearly all restaurants receiving an A grade, which restricted the models' ability to learn meaningful patterns for minority classes such as B and C. The features available in the dataset are limited and do not capture important contextual factors such as restaurant size, ownership history, or previous violations, which may strongly influence inspection outcomes. Future research can introduce additional features (such as restaurant size, ownership history, prior violations, or neighborhood-level demographic indicators) to provide the model with richer context and potentially improve its ability to detect higher-risk establishments.